

Electrical Conductivity of Mixed Electrolytes: Modeling within the Mean Spherical Approximation

Gaëlle M. Roger,^{*,†,‡} Serge Durand-Vidal,[†] Olivier Bernard,^{†,‡} and Pierre Turq[†]

UPMC Université Paris 06 and CNRS, UMR 7195, PECSA, F-75005 Paris, France

Received: March 2, 2009; Revised Manuscript Received: April 27, 2009

The purpose of this study is to predict the electrical conductivity of an electrolyte solution containing several simple ionic species. Explicit equations of the MSA-transport theory for the electrical conductivity in this complex solution are given. The theoretical conductivity of simple salts is first compared to experimental results of the literature to deduce the sizes of the ions. These sizes allow us to calculate the conductivity for a mixture of several ionic species without any additional parameter. We have also measured the electrical conductivity of solutions of LiCl, NaCl, and KCl and of KBr and MgCl₂ at 25 °C. A very good agreement between theoretical calculations and experimental values is obtained for each studied system.

1. Introduction

The computation of the electrical conductivity of concentrated aqueous electrolyte solutions is useful for many industrial processes such as batteries, plating, material transport, solid deposition, and corrosion.^{1,2}

The variation of transport coefficients of electrolytes with the concentration is one of the oldest problems in physical chemistry. In 1926³ and 1932⁴ Onsager et al. showed that, at very low concentration, the electrical conductivity is proportional to the square root of the concentration. Onsager et al. also extended this theory to higher concentrations and to mixtures in 1957⁵ using the Debye–Hückel equilibrium pair distribution functions available at this time⁶ and also by taking into account finite size corrections. The theoretical description of the electrical conductivity assumes that the solvent is a continuous medium and that ions are charged hard spheres.

The calculation of the electrical conductivity is based on the assumption that the dominant forces which determine the deviations from the ideal behavior (i.e., without any interactions between ions) are the relaxation and electrophoretic forces.⁷ The first one was introduced by Debye^{6,8,9} and appears when some external forces perturb the ionic equilibrium distribution. In that case, electrostatic forces arise which tend to restore the equilibrium distribution of ions. The second ones arise when external forces are applied to the solution: the different ions have different drift velocities. Hydrodynamic interactions of ions are mediated by the solvent, and this effect is called the electrophoretic (or hydrodynamic) effect and was initiated by Onsager.^{4,10}

In 1992¹¹ Bernard et al. proposed a new approach in which Onsager's continuity equations were combined with the MSA (mean spherical approximation) equilibrium correlation functions by means of a Green function formalism which takes the finite ionic radii into consideration. This procedure is valid up to molar concentrations.^{11–14} The MSA-transport theory, analytically solvable, allows one to describe several transport phenomena of electrolyte solutions up to molar concentrations. This approach was applied to self-diffusion coefficients,¹² acoustophoresis,^{13,15} conductivity of three

simple ionic species,¹⁶ and more complex systems such as micellar systems¹⁷ with association.¹⁸ It should also be noted that this theory has been compared with and validated by Brownian dynamics simulation several times.^{17,19,20}

The theory has been recently extended to more than three species by van Damme et al.²¹ In their work, calculated transport coefficients were compared to literature results for systems containing only two to three species. We propose a comparison between theory and experiments for electrolytes with up to four species with a better agreement.

In our work, expressions of the electrostatic relaxation corrections are close to those of van Damme et al., but we propose different expressions for the hydrodynamic corrections and the conductivity of the solution. Moreover, we compare the theoretical conductivity to experimental results for electrolytes with two to four species in solution.

We assume that the radius of the chloride ion in solution is its crystallographic one, and the value is taken in the literature. The comparison of MSA-transport calculations for electrolytes with two simple ions with experimental conductivity allows us to determine the radius of the other species in solution. These radii are then used to calculate the conductivity of electrolytes with three species or more without any additional adjustable parameters.

The paper is organized as follows. The experimental procedure is described in the next section. Then in section 3 the theoretical treatment of the electrical conductivity is presented: all formulas are explicitly given. Section 4 contains the results—including experiments and analytical calculations—and discussion.

2. Experimental Determination of the Electrical Conductivity

The conductivity experiments were performed using a Wayne–Kerr bridge, 6425 A. The conductivity cell had bright platinum electrodes. All experiments were performed at 25 ± 0.1 °C. The cell was calibrated using standard KCl solutions. The cell constant was found to be equal to 3.74 ± 0.07 cm⁻¹. This value was checked several times and did not change between the beginning and the end of the measurements. Potassium chloride (Merck, ≥99.5%), sodium chloride (Merck,

[†] UPMC Université Paris 06.

[‡] CNRS.

≥99.99%), lithium chloride (Fluka, ≥99.0%), magnesium chloride hexahydrate (VWR, ≥99.0%), and potassium bromide (VWR, ≥99.5%) were used as received. All the solutions were made up by weight. The densities of the solutions were measured with an Anton Paar DMA 38 densimeter to convert molalities into molarities.

The resistances of the solutions were measured at various frequencies (10, 5, 2, and 1 kHz) and were extrapolated by linear regressions of the resistance versus inverse frequency plots, as usual.²² The conductance results have a relative uncertainty evaluated at 1.5%. It should also be noted that we work at relatively low frequencies: this corresponds to the measure of “static” conductivity.

3. Conductance of Simple Ionic Mixtures

3.1. Presentation of the MSA-Transport Theory. As said in the Introduction, the dominant forces which determine the deviations of conductivity from the ideal behavior are the electrostatic relaxation and electrophoretic forces.⁷

In the framework of the MSA-transport theory, the expression of the electrical conductivity of an ionic solution is the following,⁴ in international units:

$$\chi = \frac{e^2}{k_B T} \sum_{i=1}^s n_i D_i^\circ z_i^2 \left(1 + \frac{\delta v_i^{\text{hyd}}}{v_i^\circ} \right) \left(1 + \frac{\delta k_i^{\text{rel}}}{k_i} \right) \quad (1)$$

where s is the number of species in the system, n_i is the number density of species i , m^{-3} , e is the elementary charge, k_B is the Boltzmann constant, T is the absolute temperature, and z_i is the valency of species i and D_i° its diffusion coefficient at infinite dilution. δv_i^{hyd} is the electrophoretic correction—also called the hydrodynamic correction—on the velocity of species i , δk_i^{rel} the relaxation correction on the electric force $k_i = z_i e E$ acting on species i , with E being the external applied electric field, and v_i° the velocity of species i due to this electric field at infinite dilution:

$$v_i^\circ = \frac{z_i e E D_i^\circ}{k_B T} \quad (2)$$

D_i° expresses the friction, which is due to the ion–solvent collisions. It corresponds to the isotonic fluctuations of the solvent, together with the so-called dielectric friction, which represents the interaction with the polarization of the surrounding solvent.^{23,24}

Ions are assumed to be charged hard spheres; in other words there is a closest distance of approach between two species. This is implied by the hard sphere repulsions. All following expressions of the corrections involve the hard sphere diameters of particles i and j , σ_i and σ_j . The closest distance of approach between ions i and j is written as σ_{ij} :

$$\sigma_{ij} = \frac{\sigma_i + \sigma_j}{2}$$

It should be noted that these diameters depend on the solvent that is taken into account: the same species can have different sizes in different solvents because of the solvation.

Explicit expressions of the correction terms are given in the following subsections.

3.2. Evaluation of the Relaxation Terms. The relaxation force $\delta \mathbf{k}_k$ is given by^{25,26}

$$\delta \mathbf{k}_k = \sum_{i=1}^s n_i \int_0^\infty \int_0^{2\pi} \int_0^\pi \nabla V_{ki} g_{ki}' r^2 \sin \phi \, d\phi \, d\theta \, dr \quad (3)$$

where V_{ki} is the effective pair potential and g_{ki}' the perturbed pair distribution function. After transformations, which are given in ref 21, this relation can be generalized and becomes

$$\frac{\delta \mathbf{k}_k}{k_k} = \frac{-\kappa^2 e_k}{3} \sum_{p=1}^s \chi_k^p \sum_{j=1}^s \sum_{i=1}^s \frac{t_j \chi_j^p \mu_i (e_i \omega_i - e_j \omega_j)}{e_i e_j (\omega_i + \omega_j)} \frac{\sinh(\kappa \sqrt{q_p} \sigma_{ij})}{\kappa \sqrt{q_p} \sigma_{ij}} \times \left[\int_{\sigma_{ij}}^\infty r \exp(-\kappa \sqrt{q_p} \sigma_{ij}) h_{ij}^\circ \, dr \right] \quad (4)$$

where $\omega_i = D_i^\circ / k_B T$, κ is the usual inverse Debye screening length

$$\kappa = \sqrt{\sum_l \frac{n_l e_l^2}{\epsilon_0 \epsilon_r k_B T}} \quad (5)$$

where e_l is the charge of species l , the relative ionic strength

$$\mu_i = \frac{n_i e_i^2}{\sum_{j=1}^s n_j e_j^2} \quad (6)$$

the mean mobility

$$\bar{\omega} = \sum_{j=1}^s \mu_j \omega_j \quad (7)$$

the transport number at infinite dilution

$$t_j = \frac{\mu_j \omega_j}{\bar{\omega}} \quad (8)$$

and

$$q_p = \sum_{i=1}^s \frac{\bar{\omega} t_i}{\omega_i + \alpha_p} \quad (9)$$

$$\chi_k^p = \frac{N_p \omega_k}{\omega_k^2 - \alpha_p^2} \quad (10)$$

where

$$\frac{1}{N_p^2} = \sum_{i=1}^s \frac{t_i \omega_i^2}{(\omega_i^2 - \alpha_p^2)^2} \quad (11)$$

and α_p is one of the s roots of the equation

$$-2\bar{\omega}\alpha \sum_{i=1}^s \frac{t_i}{\omega_k^2 - \alpha^2} = 0 \quad (12)$$

As mentioned in ref 21, Onsager demonstrated that the roots of the previous equation intertwine with the mobilities:

$$0 = \alpha_1^2 < \omega_1^2 < \alpha_2^2 < \dots < \alpha_s^2 < \omega_s^2 \quad (13)$$

We estimate the roots of eq 12 by dichotomy, and they are in very good agreement with the analytical expressions of the roots of a polynomial equation. The precision of the determination of α_p^2 by dichotomy reaches 10^{-12} .

The advantage of dichotomy is that we can simply extend the determination of the roots for several species in solution. On the contrary, the analytical determination of the roots is difficult to perform when there are more than four species in solution, i.e., when the degree of the polynomial is greater than 3.

Correcting ref 21, and by analogy with eq 14 in ref 17, the integral in eq 4 is approximated by

$$\int_{\sigma_{ij}}^{\infty} r \exp(-\kappa\sqrt{q_p}\sigma_{ij})h_{ij}^{\circ} dr \approx - \frac{e_i e_j \kappa \sqrt{q_p} \sigma_{ij} \exp(-\kappa\sqrt{q_p}\sigma_{ij})}{4\pi\epsilon_0\epsilon_r k_B T [\kappa^2 q_p + 2\Gamma\kappa\sqrt{q_p} + 2\Gamma^2 - 2\Gamma^2 Y]} \quad (14)$$

with

$$Y = \left(\frac{\sum_m^s n_m \frac{z_m^2}{(1 + \Gamma\sigma_m)^2} \exp(-\kappa\sqrt{q_p}\sigma_m)}{\sum_m^s \frac{n_m z_m^2}{(1 + \Gamma\sigma_m)^2}} \right) \quad (15)$$

These expressions are different from those in ref 21 because they take explicitly into account individual ionic sizes contrary to van Damme et al., who use individual and mean radii.

3.3. Evaluation of the Hydrodynamic Correction. The hydrodynamic (or electrophoretic) correction arises from hydrodynamic interactions between ions mediated by the solvent molecules. The expression of the correction on the velocities is the same as in Durand-Vidal et al.¹⁷ Principal equations are recalled in this paper:

$$\delta v_i^{\text{hyd}} = - \frac{eE}{3\pi\eta_o} \left(\mathcal{V}_i + \frac{\pi}{4} \sum_j n_j z_j \sigma_j^2 - \frac{\pi}{6} \sum_j n_j \sigma_j^3 \mathcal{V}_j \right) \quad (16)$$

where the last two terms take into account the asymmetry in the size of the ionic species and hard sphere interactions, and with

$$\mathcal{V}_i = \frac{\Gamma z_i}{1 + \Gamma\sigma_i} + \frac{\pi}{2\Delta} \frac{P_n \sigma_i}{(1 + \Gamma\sigma_i)} \quad (17)$$

where Γ is given in a recursive way by^{27,28}

$$4\Gamma^2 = \frac{e^2}{\epsilon_0\epsilon_r k_B T} \sum_{i=1}^s n_i \left(z_i - \frac{\pi}{2\Delta} P_n \sigma_i^2 \right) / (1 + \Gamma\sigma_i)^2 \quad (18)$$

with ϵ_0 being the permittivity of a vacuum and ϵ_r the relative permittivity of the solvent and where

$$P_n = (1/\Omega) \sum_{k=1}^s (n_k \sigma_k z_k / (1 + \Gamma\sigma_k)) \quad (19)$$

$$\Omega = 1 + (\pi/2\Delta) \sum_{k=1}^s (n_k \sigma_k^3 / (1 + \Gamma\sigma_k)) \quad (20)$$

$$\Delta = 1 - (\pi/6) \sum_{k=1}^s n_k \sigma_k^3 \quad (21)$$

Finally, the hydrodynamic correction involved in eq 1 is obtained by dividing δv_i^{hyd} by v_i° (eq 2).

4. Results and Discussion

4.1. Determination of the Radius of Each Species for Binary Electrolytes. The conductivity mainly depends on the charge, the diffusion coefficient at infinite dilution, and the closest distance of approach of the charged species that are present in solution. In this paper, the two first parameters were taken from the book of Robinson and Stokes,⁷ but the radius of the ions was deduced from the conductivity of each binary electrolyte separately. In the following, we define the radius as half of the closest distance of approach (hard sphere diameter).

The radius of the chloride ion in solution is fixed and is assumed to be equal to its crystal ionic radius: 1.81×10^{-10} m. This is the only extrathermodynamic assumption. Experimental conductivities of LiCl, NaCl, KCl, MgCl₂, and KBr were taken from the literature.^{7,29–35} The radius of the cation (Li⁺, Na⁺, K⁺, or Mg²⁺) was adjusted to fit the experimental results in a concentration range from 10^{-3} to $1 \text{ mol} \cdot \text{L}^{-1}$. The obtained radius for K⁺ was then used to deduce the radius of Br[−].

Figures representing the molar conductivity of each binary electrolyte as a function of the concentration are presented in the Supporting Information.

The radii of the different species are summarized in Table 1. It should be noted that the different presented radii are always greater than or equal to their crystallographic radii.

These sizes are in very good agreement with values found previously.¹⁶ In that previous work, the hydrodynamic and electrostatic relaxation corrections are not the same as here. On one hand, the present hydrodynamic correction is different from that of ref 16 because some terms that take into account the asymmetry of electrolytes are not neglected. On the other hand, the relaxation terms are not different from those of ref 16, but they are expressed in a different mathematical formalism. That

TABLE 1: Values of the Radii and Diffusion Coefficients of Each Species

	Li ⁺	Na ⁺	K ⁺	Mg ²⁺	Cl [−]	Br [−]
radius (Å)	0.80	1.17	1.70	1.82	1.81	1.97
D° ($10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$)	1.03	1.33	1.96	0.706	2.03	2.08

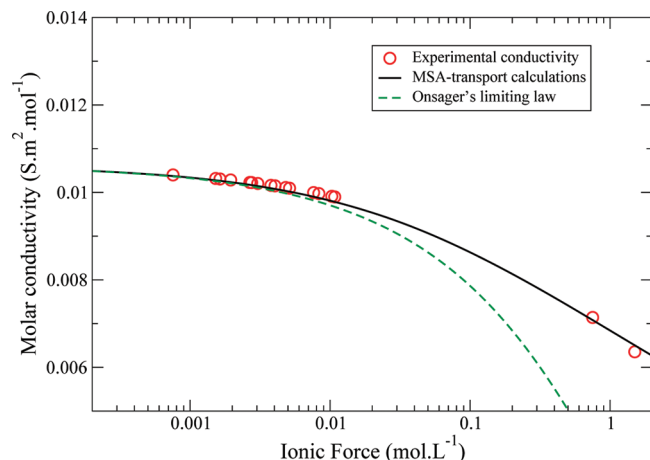


Figure 1. Molar electrical conductivity of a mixture of NaCl and MgCl₂ at 298.15 K as a function of the ionic force of the solution: experimental data (circles), MSA-transport calculations (solid line), and Onsager's limiting law (dashed line). AARD = 1.5%.

is why the sizes that we use here are not exactly the same but they are still very close.

These parameters will be used for the MSA calculations in the case of mixtures of several electrolytes. This method allows calculation of the electrical conductivity of electrolytes with more than two species without any adjustable parameter.

4.2. Comparison with Experimental Results. 4.2.1. Case of Three Simple Ionic Species. The conductivity results for a NaCl/MgCl₂ mixture in aqueous solution are presented. Experimental results are taken from the literature^{36,37} for the molar fraction in sodium X_{Na} :

$$X_{\text{Na}} = \frac{C_{\text{NaCl}}}{C_{\text{NaCl}} + C_{\text{MgCl}_2}} = 0.75$$

with C_{NaCl} the concentration in sodium chloride and C_{MgCl_2} the concentration in magnesium chloride (mol·L⁻¹).

Figure 1 presents the molar conductivity as a function of the total ionic force (I) of the solution. For convenience the molar conductivity Λ (S·m²·mol⁻¹) is used in this paper and is equal to

$$\Lambda = \frac{\chi}{I}$$

where χ is the specific conductivity (S·m⁻¹) and

$$I = \frac{1}{2} \sum_i C_i z_i^2$$

with C_i the concentration (mol·L⁻¹).

The calculated conductivity (bold line) is in good agreement with the experimental results (circles) in a concentration range between 10⁻³ and 1 mol·L⁻¹.

To evaluate objectively the difference between the experimental results and theoretical conductivity, we calculate the AARD (absolute average relative deviation):

$$\text{AARD} (\%) = \frac{100}{N} \sum_{j=1}^N \frac{|\chi_{\text{exptl}}(j) - \chi_{\text{calcd}}(j)|}{\chi_{\text{exptl}}(j)}$$

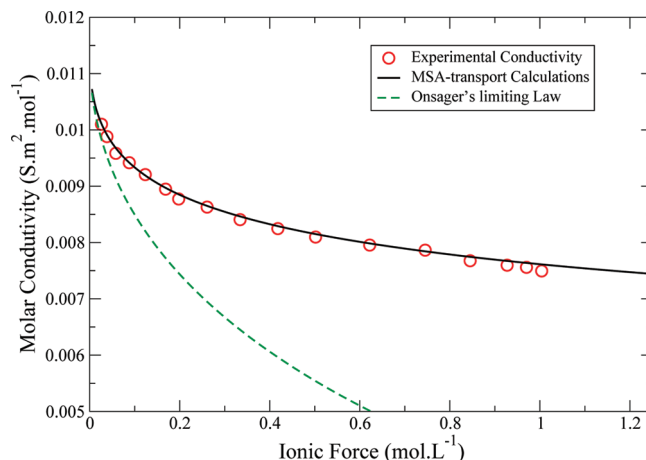


Figure 2. Molar electrical conductivity of a mixture of LiCl, NaCl, and KCl at 298.15 K as a function of the ionic force of the solution: experimental data (circles), MSA-transport calculations (solid line), and Onsager's limiting law (dashed line). AARD = 0.3%.

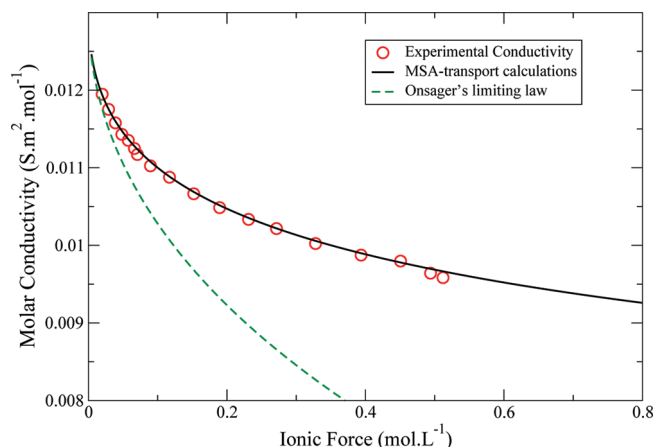


Figure 3. Molar electrical conductivity of a mixture of MgCl₂ and KBr at 298.15 K as a function of the ionic force of the solution: experimental data (circles), MSA-transport calculations (solid line), and Onsager's limiting law (dashed line). AARD = 0.6%.

where j represents the experimental points and N the number of experimental points. For this system AARD = 1.5%, which is quite low and indicates a good accordance.

Onsager's limiting law (dashed line) allows these experimental results to be reproduced up to 5×10^{-3} mol·L⁻¹.

Conductivity results of electrolyte mixtures with four different ionic species are presented in the following.

4.2.2. Case of Four Simple Ionic Species. A mixture of three 1–1 salts was first studied. We measured the conductivity of a solution with 0.118 mol/L lithium chloride, 0.142 mol/L potassium chloride, and 0.252 mol/L sodium chloride. This solution was then progressively diluted with deionized water, and we measured the conductivity for each concentration.

Figure 2 presents the molar conductivity as a function of the total ionic force (I) of the solution. MSA calculations were performed using parameters presented in Table 1.

Next, we measured the conductivity of a mixture of a 1–1 and a 2–1 salt. The solution contained initially 0.419 mol/L potassium bromide and 0.195 mol/L magnesium chloride. As previously, the solution was progressively diluted, and we measured the conductivity at different concentrations. Figure 3 presents the molar conductivity as a function of the ionic force of the solution.

For both experiments, the calculated conductivity (bold line) is in very good agreement with our experimental results (circles).

For the solutions LiCl, NaCl, and KCl and MgCl₂ and KBr, the AARD is respectively 0.3% and 0.6%, which is very low. Onsager's limiting law (dashed line) is also represented in the figures. The MSA-transport theory allows us to correctly reproduce the conductivity of simple electrolytes that have dissymmetric sizes or charges up to relatively high concentrations.

5. Conclusion

In this paper explicit equations of the MSA-transport theory in the case of several different species in solution have been presented. The theoretical calculations are in good accordance with experimental results in the case of four simple species in solution. Among the advantages of this method, the following are noteworthy: The theory provides analytical expressions that are easy to use and that can easily be extended to more than four species in solution. Moreover, once the radii of the species have been determined thanks to the conductivity of binary electrolytes, no additional adjustable parameter is needed.

This method should serve as a basis for numerous applications. Our next step will be the application of this theory to moderately concentrated nanocolloidal dispersions with added salt and the introduction of association constants between the different species in solution when it is needed. This approach provides also a way to determine physicochemical properties such as sizes or effective charges through conductivity measurements and can be extended to acoustophoresis experiments.

Acknowledgment. G.M.R. is grateful to ANDRA for financial support through Contract No. 07/0795. We gratefully acknowledge GdR PARIS.

Supporting Information Available: Figures A–E, which present respectively the molar conductivity of LiCl, NaCl, KCl, KBr, and MgCl₂ as a function of the molar concentration of salt. This information is available free of charge via the Internet at <http://pubs.acs.org>.

References and Notes

- (1) Sharygin, A. V.; Mokbel, I.; Xiao, C.; Wood, R. H. *J. Phys. Chem. B* **2001**, *105*, 229.
- (2) Hu, Y.-F.; Zhang, X.-M.; Li, J.-G.; Liang, Q.-Q. *J. Phys. Chem. B* **2008**, *112*, 15376.
- (3) Onsager, L. *Phys. Z.* **1927**, *28*, 277.
- (4) Onsager, L.; Fuoss, R. M. *J. Phys. Chem.* **1932**, *36*, 2689.
- (5) Onsager, L.; Kim, S. *J. Phys. Chem.* **1957**, *61*, 215.
- (6) Debye, P.; Hückel, E. *Phys. Z.* **1923**, *24*, 185.
- (7) Robinson, R.; Stokes, R. *Electrolyte Solutions*; Butterworths: Markham, Ontario, Canada, 1979.
- (8) Debye, P.; Falkenhagen, H. *Phys. Z.* **1928**, *29*, 121.
- (9) Debye, P.; Falkenhagen, H. *Z. Phys. Chem., A* **1928**, *137*, 399.
- (10) Onsager, L. *Ann. N. Y. Acad. Sci.* **1945**, *46*, 2689.
- (11) Bernard, O.; Kunz, W.; Turq, P.; Blum, L. *J. Phys. Chem.* **1992**, *96*, 3833.
- (12) Bernard, O.; Kunz, W.; Turq, P.; Blum, L. *J. Phys. Chem.* **1992**, *96*, 398.
- (13) Durand-Vidal, S.; Simonin, J. P.; Turq, P.; Bernard, O. *J. Phys. Chem.* **1995**, *99*, 6733.
- (14) Dufreche, J.-F.; Bernard, O.; Durand-Vidal, S.; Turq, P. *J. Phys. Chem. B* **2005**, *109*, 9873.
- (15) Durand-Vidal, S.; Turq, P.; Marang, L.; Pagnoux, C.; Rosenholm, J. *Colloids Surf., A* **2005**, *267*, 117.
- (16) Durand-Vidal, S.; Turq, P.; Bernard, O. *J. Phys. Chem.* **1996**, *100*, 17345.
- (17) Durand-Vidal, S.; Jardat, M.; Dahirel, V.; Bernard, O.; Perrigaud, K.; Turq, P. *J. Phys. Chem. B* **2006**, *110*, 15542–15547.
- (18) Roger, G. M.; Durand-Vidal, S.; Bernard, O.; Turq, P.; Perger, T.-M.; Bešter-Rogač, M. *J. Phys. Chem. B* **2008**, *112*, 16529.
- (19) Jardat, M.; Durand-Vidal, S.; Turq, P.; Kneller, G. R. *J. Mol. Liq.* **2000**, *85*, 45.
- (20) Jardat, M.; Durand-Vidal, S.; Mota, N. D.; Turq, P. *J. Chem. Phys.* **2003**, *120*, 6268.
- (21) Van Damme, S.; Deconinck, J. *J. Phys. Chem. B* **2007**, *111*, 5308.
- (22) Barthel, J.; Watcher, R.; Gores, H. *J. Modern Aspects of Electrochemistry*; Plenum Press: New York, 1979; Vol. 13.
- (23) Chandra, A.; Bagchi, B. *J. Chem. Phys.* **1999**, *110*, 10024.
- (24) Biswas, R.; Bagchi, B. *J. Am. Chem. Soc.* **1997**, *119*, 5946.
- (25) Résibois, P. M. V. *Electrolyte Theory*; Harper and Row: New York, 1968.
- (26) Ebeling, W.; Feistel, R.; Kelbg, G.; Sändig, R. *J. Non-Equilib. Thermodyn.* **1978**, *3*, 11.
- (27) Blum, L. *J. Mol. Phys.* **1975**, *30*, 1529.
- (28) Blum, L.; Høye, J. *J. Phys. Chem.* **1977**, *81*, 1311.
- (29) Lobo, V. M. M.; Quaresma, J. L. *Electrolyte Solutions: Literature Data on Thermodynamic and Transport Properties*; University of Coimbra: Coimbra, Portugal, 1981; Vol. 2.
- (30) Shedlovsky, T. *J. Am. Chem. Soc.* **1932**, *54*, 1411.
- (31) Chambers, J. F.; Stokes, J. M.; Stokes, R. H. *J. Phys. Chem.* **1956**, *60*, 985.
- (32) Carman, P. C. *J. Phys. Chem.* **1969**, *73*, 1095.
- (33) Miller, D. G. *J. Phys. Chem.* **1966**, *70*, 2639.
- (34) Swain, C. G.; Evans, D. F. *J. Am. Chem. Soc.* **1966**, *88*, 383.
- (35) Shedlovsky, T.; Brown, A. S. *J. Am. Chem. Soc.* **1934**, *56*, 1066.
- (36) Bianchi, H.; Corti, H. R.; Fernández-Prini, R. *J. Solution Chem.* **1989**, *18*, 485.
- (37) Bianchi, H.; Corti, H. R.; Fernández-Prini, R. *J. Solution Chem.* **1992**, *21*, 1107.

JP901916R